Fields

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1 Character of a field

1. Morphism between fields must be injective.
2. Character of a field must be prime or 0.
3. If the character is 0, the field contains a copy of $\mathbb{Q}$.
4. If the character is a prime p, the field contains a copy of $\mathbb{F}_p$.
5. If there is morphism from F to K, then they must have the same character.

   > **Remark 1.1.** Fields with different characters are basically unrelated.

2 Field Extensions

1. Every field is an extension of its prime field or $\mathbb{Q}$.
2. $k \subseteq F$. F is a vector space over k and moreover a k-algebra.
3. The degree of extension $[F : k]$ is the dimension of F as a vector space over k.
4. If $k \subseteq F \subseteq K$, then $[K : k] = [K : F][F : k]$.

3 Simple Extensions

1. An extension F/k is simple if $F = k(\alpha)$ for some $\alpha \in F$. $k(\alpha)$ is the smallest subfield of F containing k and α.
2. Examples of simple extensions: $\mathbb{Q}(\sqrt{2})$, $\mathbb{F}_p(t)$. Is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ simple? Yes, $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$ (look at minimal polynomial of $\sqrt{2} + \sqrt{3}$ over $\mathbb{Q}$ and the degree of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$).

3. Surprisingly, **every finite extension of character 0 is simple !**. In fact, it's very hard to find an example of non-simple extension.

4. So what does $k(\alpha)$ look like? Consider the evaluation morphism $f : k[x] \to F$ defined by $f(p(x)) = p(\alpha)$. The kernel of this morphism is an ideal in $k[x]$, and since $k[x]$ is a principal ideal domain, the kernel is generated by a single polynomial, called the minimal polynomial of α over k, and we say α is algebraic over k. If the minimal polynomial is 0, then $k(\alpha)$ is isomorphic to the field of rational functions $k(x)$.

5. The degree of the minimal polynomial of α over k is equal to $[k(\alpha) : k]$, why?

4 Algebraic Number

If $F = k(\alpha_1, \alpha_2, \dots, \alpha_n)$ (finitely generated), then the following are equivalent:

1. $[F : k]$ is finite.

2. F is made of entirely algebraic elements over k.

3. Each α_i is algebraic over k.

Key of proof: $k \subseteq k(\alpha_1) \subseteq k(\alpha_1, \alpha_2) \subseteq \cdots \subseteq k(\alpha_1, \dots, \alpha_n) = F$

Remark 4.1. This is very powerful.

- If a and b are algebraic over k, then $a + b$, $a - b$, ab, and a/b (if $b \neq 0$) are also algebraic over k.

- Let E be the set of algebraic numbers over $k \subseteq F$, then E is a field. Furthermore, E is an extension of k called the algebraic closure of k in F (why?).

5 splitting field

1. Unique up to isomorphism. We skip the proof.

2. Splitting field must be a finite extension.

3. Example: splitting field of $x^4 + 1$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2}, i)$.

4. Example: splitting field of $x^4 + 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[4]{2}, i)$.